

AGX CRM

Detailed Product Requirements & Business Operating System Specification

Internal CRM + Sales + Client Management + Project Operations +
Finance + Productivity + Analytics

Document	AGX CRM Master Product Specification
Version	1.0
Prepared For	AGX / AGXperience
Primary Objective	Create a single source of truth for business operations
Recommended Stack	React / Next.js + Supabase + PostgreSQL
Future Direction	AI-powered Business Operating System + Client Portal

Confidential — Internal Product & Development Specification

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1. Product Vision & Objectives

AGX CRM is intended to become the central operating system for AGX. It should replace fragmented workflows across spreadsheets, WhatsApp, email, cloud folders, personal notes and disconnected tools.

Core lifecycle: Lead → Client → Agreement → Payment → Project → Tasks → Team → Delivery → Profitability → Client History

The system must provide a single, permission-controlled source of truth for sales, clients, projects, money, employees, documents, credentials and operational performance.

Primary objectives:

- Manage every lead systematically from acquisition through conversion.
- Convert qualified leads into clients without duplicate data entry.
- Track every client project, milestone, task, deadline and owner.
- Track agreements, invoices, payments, outstanding amounts and expenses.
- Calculate business and project-level revenue, cost, profit and margin.
- Manage multiple employees with role-based permissions.
- Measure productivity through meaningful output and delivery metrics.
- Securely store project credentials and sensitive documents.
- Maintain a complete audit trail of important business actions.
- Provide actionable dashboards that help management make decisions quickly.

2. Website Integration & Authentication

The public website should remain customer-facing. The CRM should be accessible through a discreet Admin Login entry in the header or footer.

Route	Purpose	Access
/admin/login	Authentication screen	Public
/admin/dashboard	CRM command center	Authenticated users
/admin/leads	Lead management	Permission controlled
/admin/clients	Client management	Permission controlled
/admin/projects	Project management	Permission controlled
/admin/tasks	Tasks and productivity	Permission controlled
/admin/finance	Revenue, payments, expenses	Finance permission
/admin/analytics	Business analytics	Management permission
/admin/settings	System configuration	Admin permission

Login requirements:

- Email/username and password authentication.

- Show/hide password control.
- Forgot-password flow.
- Remember-me option where appropriate.
- Secure session handling and logout.
- Optional two-factor authentication in a later phase.
- All CRM routes must enforce authentication and authorization at the backend/database layer.

3. User Roles & Permission Model

The CRM must use role-based access control (RBAC). Permissions must be enforced server-side and at the database layer, not only by hiding buttons in the interface.

Role	Primary Access	Key Restrictions
Super Admin	Entire system	None except platform-level safeguards
Admin	Most business operations	Limited security/system controls
Sales Manager	Leads, clients, pipeline, sales reports	No unnecessary credentials/system controls
Sales Executive	Assigned leads, clients, follow-ups	No company-wide finance or sensitive credentials
Project Manager	Projects, milestones, tasks, team	No unrelated finance/security data
Developer / Employee	Assigned projects and tasks	Only relevant client/project data
Accountant	Payments, invoices, expenses, financial reports	No unnecessary project credentials

The system should support custom roles in a later phase so management can define granular permissions such as view/create/edit/delete/export/reveal-credential.

4. Dashboard & Command Center

The dashboard is the first screen after login and must communicate the most important business information within approximately 5–10 seconds.

KPI cards:

Area	KPIs
Sales	Total leads, new leads, qualified leads, conversion rate, won deals, lost deals
Clients	Total clients, active clients, new clients, inactive clients
Projects	Active, completed, delayed, due soon
Finance	Revenue, expenses, net profit, outstanding payments, profit margin
Productivity	Open tasks, completed tasks, overdue tasks, team completion rate

Core visualizations:

- Revenue vs expenses by month.
- Lead funnel from new lead to won deal.
- Project status distribution.

- Outstanding payment list.
- Upcoming task and project deadlines.
- Team productivity snapshot.

Quick actions should include: Add Lead, Add Client, Add Project, Create Invoice, Record Payment, Add Expense and Create Task.

5. Lead Management

Lead management is the sales engine of the CRM.

Lead fields:

Field Group	Fields
Identity	Lead ID, name, company, phone, email, WhatsApp, location
Commercial	Interested service, estimated deal value, probability
Source	Website, WhatsApp, Instagram, Facebook, Google, LinkedIn, referral, cold call, advertisement, other
Operations	Assigned salesperson, priority, status, next follow-up, last contacted
Context	Notes, documents, activities, created date

Pipeline:

NEW → CONTACTED → QUALIFIED → PROPOSAL SENT → NEGOTIATION → WON / LOST

The pipeline should support Kanban drag-and-drop. Status changes must create an activity/audit event.

6. Client Management & 360° Client View

When a lead is marked Won, the user should be able to convert it to a client without re-entering existing information.

Client profile should contain:

- Client ID, person/company name, phone, email, website and address.
- Industry, GST/tax information and contact persons.
- Assigned account manager.
- Total business value, total payments and outstanding amount.
- Number and status of projects.
- Client acquisition source and relationship notes.

Client 360° tabs:

Overview | Projects | Payments | Invoices | Agreements | Documents | Tasks | Credentials | Communication | Activity | Notes

Every client record should provide a chronological timeline showing the relationship from lead acquisition through delivery and ongoing maintenance.

7. Project Management

Projects are the operational core for AGX's software, website, mobile app and automation work. Each client can have multiple projects.

Project fields:

Field	Purpose
Project ID / Name	Unique project identification
Client / Service Type	Connect project to customer and offering
Project Manager	Accountable owner
Assigned Team	People responsible for delivery
Dates	Start, expected delivery, actual delivery
Commercials	Project value, received, pending
Status / Priority	Operational state and urgency
Progress	Overall completion percentage
Description	Scope and context

Default project workflow:

NOT STARTED → PLANNING → IN PROGRESS → TESTING → CLIENT REVIEW → REVISION → COMPLETED → MAINTENANCE

Progress should preferably be calculated from milestone/task completion instead of being an arbitrary manual number.

8. Task & Productivity Management

Tasks can belong to a client, project, employee or department.

Task fields:

- Task ID and title.
- Description and attachments.
- Client and project.
- Assigned user.
- Priority and status.
- Start date and due date.
- Estimated hours and actual hours.
- Comments and completion history.

Statuses: TO DO → IN PROGRESS → IN REVIEW → COMPLETED, with BLOCKED as an additional state.

Task views:

- List view

- Kanban board
- Calendar view
- My Tasks view

Productivity should focus on output and delivery rather than superficial activity. Metrics include tasks completed, overdue tasks, on-time completion, estimated vs actual hours and project contribution.

Optional productivity score:

Productivity Score = weighted combination of task completion, on-time delivery, project contribution and overdue-task performance. The weighting must be configurable.

9. Finance, Payments & Profitability

The finance module must provide a reliable operational view of AGX's money without pretending to replace a full statutory accounting system.

Finance navigation:

Finance → Revenue | Payments | Invoices | Expenses | Profit & Loss

Revenue/payment fields:

- Payment ID, client, project, invoice, amount and payment date.
- Payment method, transaction/reference ID and notes.
- Payment status: Pending, Partially Paid, Paid, Overdue, Refunded.

Payment milestones should support structures such as 50% advance + 50% completion or 30% + 40% + 30%. The system calculates received amount, outstanding amount and payment percentage automatically.

Expense fields:

- Expense ID, category, amount, date, vendor and description.
- Payment method and receipt attachment.
- Optional client/project relationship.
- Created by and approval status where required.

Core financial formula:

Net Profit = Revenue – Expenses

Project profitability should also account for direct project expenses and, where desired, allocated employee cost.

Example: Project Revenue ₹2,00,000 – Developer Cost ₹40,000 – Tools ₹10,000 – Other Expenses ₹5,000 = Project Profit ₹1,45,000.

10. Agreements & Document Management

Each client/project should have a dedicated document workspace.

Document types:

- Agreements / contracts
- NDAs
- Proposals
- Quotations
- Invoices
- Payment receipts
- Requirements
- Designs and project files
- Other supporting documents

Agreement fields:

- Agreement name/type.
- Client and project.
- Start and expiry date.
- Commercial value.
- Status: Draft, Sent, Signed, Expired, Cancelled.
- Signed document and signature status.

11. Secure Credentials Vault

The credentials vault is a high-risk module because it may contain passwords, API keys and access to client infrastructure.

Credential record:

- Credential name and platform.
- URL, username and password.
- API key/token where applicable.
- Client and project.
- Notes.

Security requirements:

- Encrypt sensitive credential values at rest.
- Never display secrets by default.
- Show masked values such as
- Require appropriate permission to reveal/copy.
- Log sensitive actions such as reveal/copy when feasible.
- Do not expose credentials in normal list APIs or client-side source.

12. Calendar & Notifications

Central calendar should combine meetings, follow-ups, task deadlines, project deadlines, payment reminders and agreement expiry dates.

Notification examples:

- Payment overdue
- Task due tomorrow
- New lead assigned
- Project deadline approaching
- Agreement expiring
- Payment received
- New task assigned

Notification preferences should be configurable per user and event type.

13. Business Analytics & Reporting

Analytics should help management understand what is driving growth, where money is being made, and where operations are failing.

Analytics Area	Metrics
Sales	Leads, sources, conversion rate, average deal value, sales cycle, won/lost
Clients	New clients, repeat clients, retention, lifetime value, revenue/client
Projects	Completion rate, delays, delivery time, profitability
Finance	Revenue, expenses, profit, margin, outstanding, cash-flow view
Team	Tasks completed, on-time delivery, workload, project contribution
Services	Revenue, cost, profit and margin by service type

Reports should support monthly, quarterly, yearly and custom date-range filtering.

14. Audit Logs & Activity Tracking

Every important operational action should be traceable.

Audit record should capture:

- User who performed the action.
- Action type.
- Entity and entity ID.
- Timestamp.
- Relevant before/after values where appropriate.
- IP/device information only where justified by security requirements.

Example: 20 Aug 2026 — Abhinav — Updated project status — ABC CRM: Development → Testing.

15. Global Search

Provide one universal search field across the CRM.

Search should return relevant records from leads, clients, projects, tasks, payments, documents and agreements.

Example query: Shrawello → Client profile, active projects, payments, agreements, documents and related tasks.

Search should support partial matches, filters and permission-aware results so users never discover records they are not authorized to view.

16. Recommended Navigation & Page Architecture

Primary navigation:

- AGX CRM
 - Dashboard
 - Leads
 - Clients
 - Projects
 - Tasks
 - Calendar
 - Finance
 - Revenue
 - Payments
 - Invoices
 - Expenses
 - Profit & Loss
 - Documents
 - Credentials
 - Team
 - Analytics
 - Notifications
 - Global Search
 - Audit Logs
 - ⚙ Settings

Each module should have: list view, filters, search, create action, detail view, edit action and permission-aware actions wherever relevant.

17. Database Architecture

Recommended relational core entities:

- users
- roles
- permissions
- leads
- lead_activities
- clients
- client_contacts
- projects
- project_members
- project_milestones

tasks
task_comments
task_attachments
invoices
payments
expenses
agreements
documents
credentials
notifications
calendar_events
activity_logs
audit_logs

Key relationships:

LEAD → CLIENT → PROJECT → MILESTONES → TASKS → TEAM MEMBERS

CLIENT → INVOICES → PAYMENTS

CLIENT → AGREEMENTS → DOCUMENTS

PROJECT → CREDENTIALS

The data model should avoid duplicating business-critical information. Financial records should be immutable enough to preserve history, with adjustments/reversals rather than destructive edits where appropriate.

18. Core Business Workflows

Lead-to-client workflow:

LEAD → QUALIFICATION → PROPOSAL → NEGOTIATION → WON → CLIENT

Client-to-project workflow:

CLIENT → AGREEMENT → PAYMENT → PROJECT → MILESTONES → TASKS → DELIVERY → MAINTENANCE

Payment workflow:

INVOICE → PAYMENT RECEIVED → BALANCE UPDATED → PROJECT/CLIENT FINANCIAL HISTORY UPDATED

Project workflow:

PLANNING → DEVELOPMENT → TESTING → CLIENT REVIEW → REVISION → COMPLETED → MAINTENANCE

19. Security & Data Protection

Security must be treated as a first-class product requirement.

- Use secure authentication and session management.
- Enforce role-based authorization on backend APIs/database policies.
- Use database Row Level Security where supported.
- Encrypt sensitive credentials at rest.
- Use HTTPS in production.
- Restrict document storage access through authenticated policies.

- Use secure password hashing through the authentication provider.
- Implement database backups and recovery procedures.
- Maintain audit logs for sensitive and financial actions.
- Never expose service-role or privileged database credentials in frontend code.
- Apply least-privilege access by default.

Credential data should be treated separately from ordinary CRM data and should have stronger access controls.

20. UI/UX Requirements

The CRM should feel like a modern premium SaaS platform rather than an outdated ERP.

- Clean, minimal interface.
- Fast navigation and low interaction friction.
- Clear typography and strong hierarchy.
- Responsive desktop-first dashboard with mobile support.
- Light and dark themes if practical.
- Data-dense tables with useful filters rather than excessive decoration.
- Consistent status badges and priority indicators.
- Clear empty states and helpful onboarding.
- Confirmation for destructive actions.
- Toast notifications for successful/failed operations.
- Keyboard-friendly search and navigation where feasible.

The dashboard should prioritize information that changes decisions. Avoid charts simply for visual decoration.

21. Technical Architecture

Recommended initial stack:

Layer	Recommendation
Frontend	React or Next.js
UI	Modern component system + responsive design
Backend	Supabase services / server-side API layer
Database	PostgreSQL
Authentication	Supabase Auth or equivalent secure identity provider
Storage	Supabase Storage or equivalent private object storage
Authorization	RBAC + PostgreSQL Row Level Security
Deployment	Netlify/Vercel or equivalent
Analytics	SQL-based reporting + charting layer

Architecture should keep business logic, authorization and sensitive operations away from the browser. The frontend is a presentation layer, not the security boundary.

22. MVP, Phase 2 & Phase 3 Roadmap

Phase 1 — Core CRM

- Authentication and user roles
- Dashboard
- Leads and pipeline
- Clients
- Projects
- Tasks
- Payments
- Expenses
- Documents
- Basic analytics

Phase 2 — Business OS

- Advanced finance
- Project profitability
- Productivity analytics
- Agreements
- Credentials vault
- Audit logs
- Calendar
- Notifications
- Advanced reports

Phase 3 — Intelligence

- AI business assistant
- Automated lead follow-ups
- WhatsApp/email integrations
- Client portal
- Automated management reports
- AI project risk detection
- AI sales forecasting

23. Acceptance Criteria

The first production release should satisfy these conditions:

- 1 Unauthorized users cannot access CRM data or pages.
- 2 Admins can create and manage multiple users with different roles.
- 3 A lead can move through a controlled sales pipeline.
- 4 A Won lead can be converted into a client without duplicate entry.
- 5 Clients can have multiple projects.
- 6 Projects can have milestones, team members and tasks.
- 7 Tasks can be assigned, tracked and completed.
- 8 Payments automatically update client/project outstanding balances.
- 9 Expenses are reflected in financial reporting.
- 10 Revenue, expenses, net profit and margin can be calculated for selected date ranges.
- 11 Documents can be attached to the correct client/project.
- 12 Credentials are masked and permission controlled.
- 13 Important changes create audit/activity records.
- 14 Analytics respect the logged-in user's permissions.
- 15 Data can be searched quickly across relevant CRM modules.

24. Future AI & Automation Layer

AI should be added on top of structured operational data, not used as a substitute for good database design.

Potential commands:

- "How much revenue did we generate this quarter?"
- "Which clients currently owe us money?"
- "Which projects are likely to be delayed?"
- "Which service generated the highest profit?"
- "Show me all high-value leads without a follow-up in the last 7 days."
- "Summarize today's operational priorities."

Automation opportunities:

- Website lead → CRM → salesperson assignment → notification → follow-up.
- Payment overdue → internal alert → customer reminder.
- New project → standard task template creation.
- Deadline approaching → project manager notification.
- Agreement expiry approaching → renewal reminder.
- Weekly management report → automatically generated summary.

25. Final Product Vision

The long-term product should evolve from a CRM into an AGX Business OS.

The platform should connect four operating layers:

Layer	Capabilities
Sales	Leads, pipeline, clients, proposals, follow-ups
Operations	Projects, milestones, tasks, team, delivery
Finance	Revenue, payments, expenses, profitability
Intelligence	Analytics, automation, AI assistant, forecasting

Future client portal:

- Project progress
- Milestones
- Documents
- Agreements
- Invoices
- Payment history
- Outstanding balance
- Project updates
- Support requests

Clients must never see internal notes, internal costs, employee-private information or credentials.

The strategic objective is simple: one system that tells AGX what is happening, who is responsible, what money is moving, what is profitable, and what requires attention next.

Implementation Principles

- Build the data model before polishing the UI. Correct relationships are the foundation.
- Security before convenience. Credentials, financial records and documents require strict authorization.
- Automate repetitive operations. Do not force staff to manually update the same information in multiple places.
- Track history. Business systems need accountability and auditability.
- Design for scale. The architecture should support more users, clients, projects and future integrations.
- Measure profitability, not just revenue. Revenue without cost visibility can produce misleading growth.
- Make the dashboard decision-oriented. The goal is faster, better management decisions.

End of Product Specification